

ARDUINO PHYSICAL AI CHALLENGE INDIA 2026

Project Report

Organised by Robu.in × Arduino · Submit as PDF · Max 10 MB

Project Title	AutoFrame: An AI-Powered Hands-Free Video Call Assistant
Team Name	PIDTuners
Registration / Team ID	APC-2026-PY12546
Contest Track	Smart Home and Consumer AI
Institution & City	National Institute of Technology Puducherry, Karaikal

Team Members (1–4)

Name	Email	Role
Anmol Sinha	ee24b1006@nitpy.ac.in	Team Lead
Achyut Kuumar	achyutkuumar67@gmail.com	Member
Aaron D	ee24b1002@nitpy.ac.in	Member
Shresht K	shreshtnadadur@gmail.com	Member

1. Project Overview

Problem Statement

What real-world problem does your project solve? Who faces it? 2–5 sentences.

During video calls, users often struggle to remain in frame when moving around their desk, giving a presentation, or holding up objects. To stay visible, they must constantly interrupt their workflow to manually adjust their webcam. This project solves this by creating a hands-free, motorized gimbal that automatically tracks the user's face keeping them perfectly framed at all times without manual intervention.

How Your Project Works

The system captures a live 720p video feed using a high-definition webcam, sending it directly to the Arduino UNO Q via a USB connection. Utilizing a lightweight YOLO edge AI model, the onboard Qualcomm Linux processor actively detects the primary subject in the frame and calculates their precise X and Y bounding box coordinates in real-time. These target coordinates are then seamlessly transmitted internally to the secondary Cortex-M33 microcontroller. Instead of relying on a custom control loop, the microcontroller leverages the AccelStepper library to dynamically compute the necessary step trajectories and smooth

acceleration profiles. This data is efficiently converted into precise step and direction signals to drive the two NEMA 17 stepper motors, seamlessly panning and tilting the camera bracket. Finally, a pre-programmed deadzone buffer ensures the camera only actuates when the subject steps significantly out of the centre frame.

Why Arduino UNO Q?

How does its on-device AI capability enable your solution?

The UNO Q's dual-brain architecture perfectly isolates our two heaviest workloads. The Qualcomm Linux processor natively handles the complex AI computer vision (YOLO) and USB video ingestion without lag. Simultaneously, the Cortex-M33 microcontroller manages the strict, real-time closed-loop PID math required for smooth motor actuation.

2. Components Used (BOM)

List every component. The Arduino UNO Q purchase proof must be uploaded separately with your submission.

Component	Qty
Arduino UNO Q (ABX00087)	1
Logitech C270 HD Webcam	1
NEMA 17 Stepper Motors	2
A4988 Stepper Motor Drivers	2
12V DC Wall Adapter (3A)	1
Electrolytic Capacitors (470µF, 25V)	1
Pan-Tilt Camera Bracket & Mounting Hardware	1

3. System Architecture & Circuit

Step-by-Step Workflow

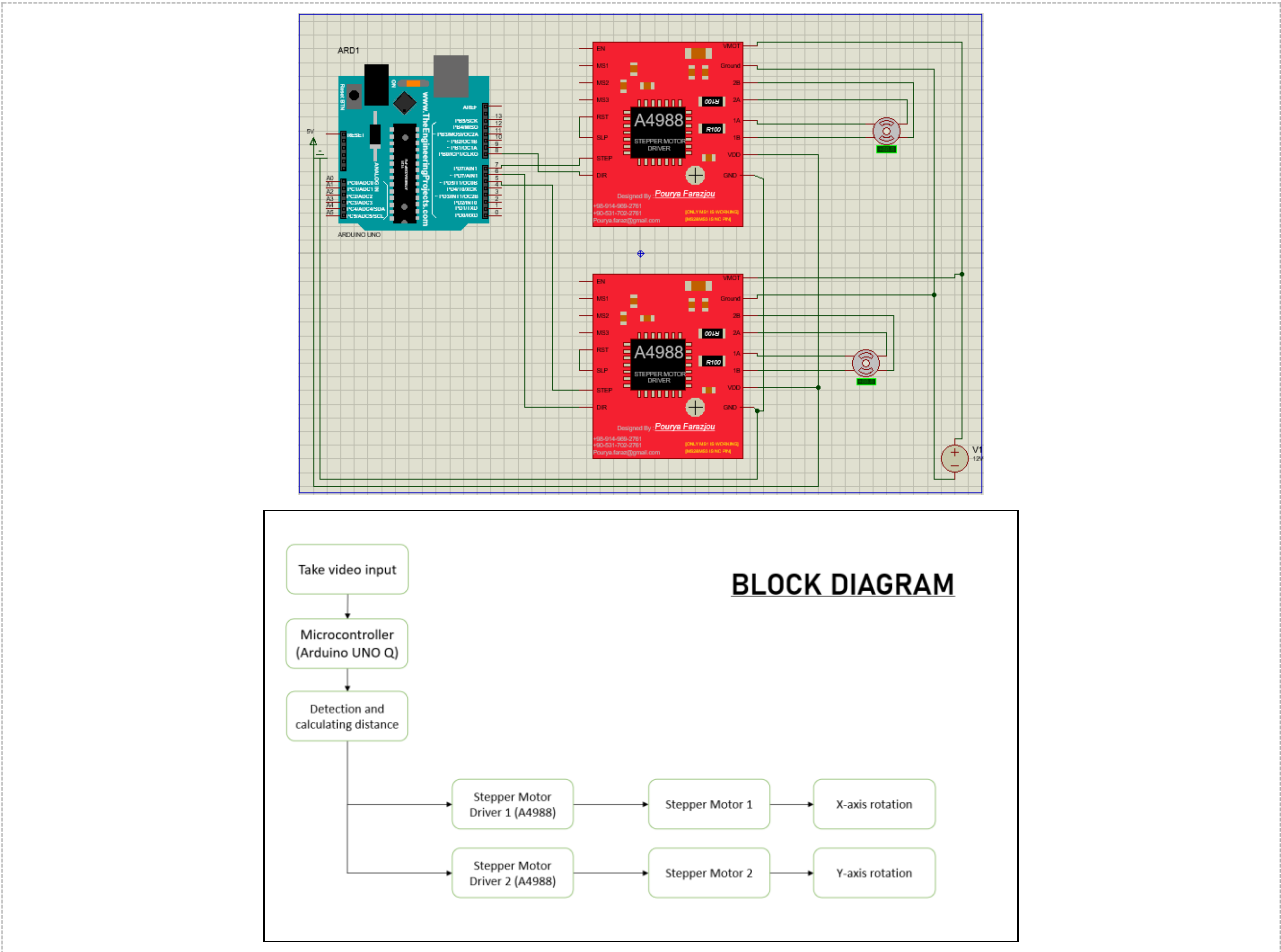
Example: Sensor collects data → UNO Q runs AI model on-device → Output device acts.

Webcam captures video feed via USB → Qualcomm Linux processor on the UNO Q runs a YOLO AI model locally to calculate target X and Y bounding box coordinates → The Qualcomm processor transmits the target coordinates internally to the Cortex-M33 microcontroller → The

Cortex-M33 evaluates deadzone buffer logic and calculates smooth movement trajectories using the AccelStepper library → A4988 drivers actuate NEMA 17 stepper motors to physically pan and tilt the camera bracket.

Block Diagram & Circuit Schematic

Paste your block diagram and circuit/wiring diagram below. Hand-drawn photos, Fritzing, KiCad — all accepted. Label pin connections.



4. AI / ML Model Details

If your project does not use AI/ML, write “Not Applicable” — but note that Physical AI projects score higher on Innovation.

Model Used	MobileNetV2 SSD
------------	-----------------

Training Platform	TensorFlow Lite
Accuracy	N/A - Standard Pre-trained Model
Dataset	COCO Dataset

Brief Description & Limitations

How does the model work in your project? When does it fail or degrade?

The Qualcomm Linux processor locally runs a lightweight YOLO model on a 720p feed to rapidly extract X and Y bounding box coordinates without bottlenecking the system's memory.

Limitations: The model's constraints primarily appear in complex framing scenarios when multiple faces are detected.

5. Code Structure

Briefly explain the main functions. Example: `setup()` → initialises sensors; `loop()` → main execution; `detectObject()` → detection logic.

1. Python (main.py): `on_all_detections(detections)` → calculates X and Y tracking errors, applies deadzone filtering, and sends calculated motor steps to the microcontroller via Bridge
2. Arduino (sketch.ino): `pan_camera(steps)` → receives Bridge input and queues the target step distance for the X-axis (pan) motor; `tilt_camera(steps)` → receives Bridge input and queues the target step distance for the Y-axis (tilt) motor; `setup()` → initializes variables, bridge communication; `loop()` → updates the Bridge listener for incoming Python commands and continuously executes the non-blocking stepper motor movements.

GitHub Repository	https://github.com/AnmolSinha42/VideoCallAdjuster
Demo Video Link	https://drive.google.com/drive/folders/12pTF1gvld_nGMitL37l93FxTXB-Zv1_n?usp=drive_link

6. Testing & Results

How did you test the complete system? What were the results? Include accuracy, response time, edge cases.

The facial recognition was tested on multiple different faces and it works as expected.

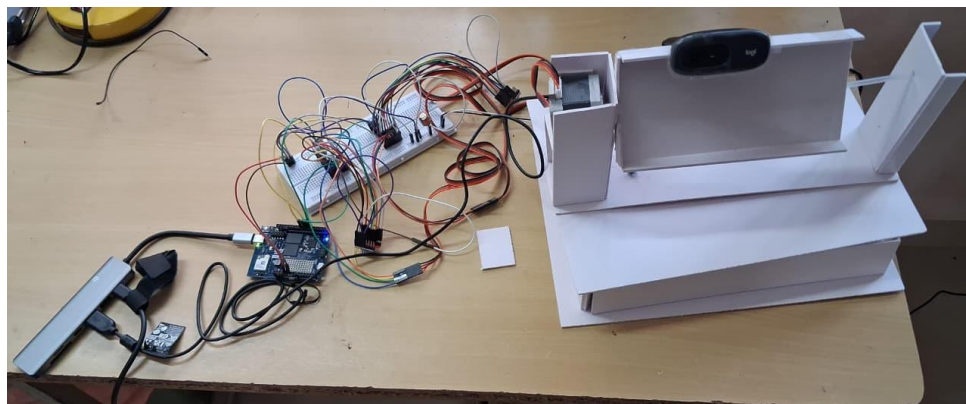
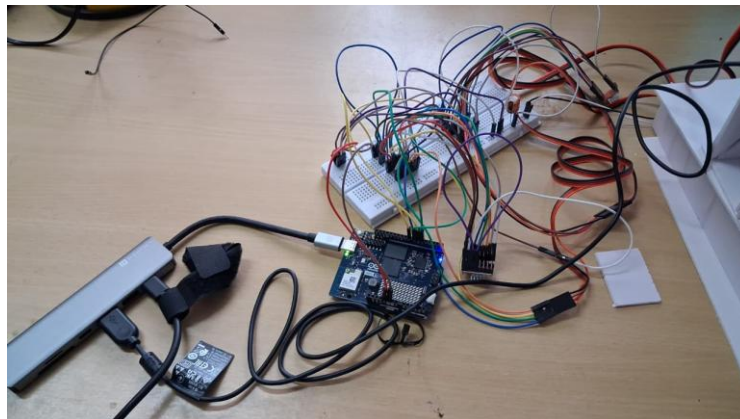
Edge Case: the project is only designed for small, slow and natural movements (such as standing up or down) or just trying to reach/show something. In other cases where the user quickly walks away, it is unable to keep up with the user.

Response time = 0.1s

Accuracy is not required as the user need not be exactly in the centre of the frame for video calling applications.

Project Images (2–3)

[Insert 2–3 photos of your project working]



7. Challenges, Learnings & Future Improvements

Challenges Faced

e.g. sensor calibration, model optimisation, power management.

1. The rotor shaft and the object attached to it were slipping causing issues.
2. the A4988 chips are prone to thermal throttling; if the 600g payload demands too much current, the drivers will overheat and enter thermal shutdown unless the reference voltage (VREF) is perfectly capped at around 1.2V.

What You Learned & What's Next

Summarise what you built, what you learned, and 2–3 future improvements.

We successfully built a dual-brain physical AI system, bridging high-level computer vision on Debian Linux with low-level closed-loop electrical control on a Cortex-M33. We learned how to manage power architecture, specifically isolating the 12V motor supply from the logic board while utilizing 100 μ F - 220 μ F capacitors to absorb back-EMF voltage spikes.

Future Improvements:

1. Having the body of the gimbal 3D printed
2. Using TMC2209 stepper motor drivers rather than the A4988.

8. Declaration

We confirm this is our original, unpublished work. The Arduino® UNO™ Q is the primary board in this project. All team members have reviewed and agree to this report.

Date	23 August 2026
------	----------------